

HOW LOW CAN YOU GO?

Evolving FPGAs tackle low-power demands in the increasingly mobile world

■ WENDY LOCKHART

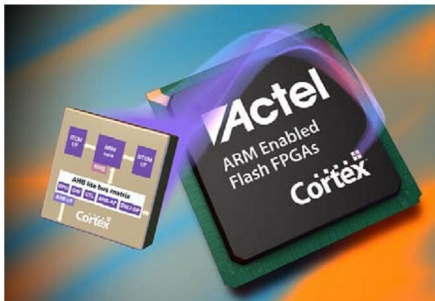
The tremendous growth in battery-operated and power-sensitive applications—not to mention the fast-rising design-for-energy-efficiency dynamic—has created a global demand for low-power semiconductors. Indeed, it's one of the industry's driving factors today, and it's putting enormous pressure on system engineers who face stricter design specifications and standards that cap total system power consumption. A few extra milliwatts or even microwatts can add up to a major power-budget headache for a portable device designer.

But at the same time, demand continues unabated for more features, performance and flexibility. These needs must be met without draining the battery or increasing costs. Managing these design tradeoffs has become a paramount concern for today's system engineers—in every discipline from mobile phones to medical equipment.

Searching for answers

To deal with this, designers have relied on application-specific integrated circuits (ASICs), not FPGAs, to meet their low-power constraints. But ASICs can hamstring engineers: Using them means longer time-to-market, hefty non-recurring engineering charges and a lack of flexibility to address changing standards and late-stage design modifications. Thus hard-wired ASICs are riskier and often impractical for applications with short product lifecycles.

Similarly, complex programmable logic devices (CPLDs) used in some low-power applications are losing their effectiveness, due to relatively high costs and the increased demand for high-end features and extra logic. CPLDs do not



FPGA-optimised 32-bit ARM Cortex-M1 processor

offer the level of integration, flexibility or sophistication required for most of today's applications.

Enter FPGAs. More and more designers are looking to a low-power, reprogrammable solution so that they can adapt to evolving standards, speed up time-to-market, and deliver the footprint and power consumption required. This shifting interest is so keen that market research firm iSuppli predicts as much as \$3 billion of the estimated \$20 billion ASIC market could potentially shift to low-power FPGA solutions in the coming years.

But not all programmable logic is suited to address low-power needs. FPGAs based on static access random memory (SRAM) technology have inherently high static power consumption. Even 'low-power' SRAM-based FPGAs draw of the order of ten times more power than specified for typical battery-operated applications. Some of these 'low-power' FPGAs draw up-

wards of 30 mA, which is often one or two orders of magnitude higher than typical power-sensitive, battery-operated applications can tolerate. At start-up, SRAM-based devices experience well-documented inrush and boot-up configuration power spikes during system initialisation that can cause excess battery drain and cause failures.

Power problems

Once the FPGA is 'on' and configured, power consumption takes two basic forms—static and dynamic (also called 'active'). Static power consumption is the current drawn by an FPGA when it is powered up, configured and doing nothing, while dynamic power is consumed when devices are actively working.

Until recently, dynamic power was the dominant source of power consumption. Once helping to manage the dynamic power problem, device supply voltages (V_{cc}) had dropped with process shrinks and subsequent lower